

Data Proposal

Item-response data for measuring ability and tracking its change

Research context

We are developing a measurement model that estimates a respondent's ability from a history of item responses and tracks how that ability changes over time. It places different response formats on a single common scale and can link new items onto an existing scale without a full recalibration. We have evaluated it on simulated and public data, and we are now seeking real response data to test it at scale. We would value whatever can be shared, suitably anonymized and under an agreement that works for you.

Data of interest

Listed roughly in order of value to us, and any single category is useful on its own. In every case a respondent identifier, an item or test identifier, a score, and a date or sequence index are usually enough to begin. Richer detail is welcome but not required.

1. **Longitudinal response data.** Repeated measurements of the same respondents across occasions, terms, or years. This is the most valuable category, since it lets us estimate ability growth and its rate.
2. **Item-level responses.** Scored responses at the item level rather than test totals, with a reasonable number of respondents per item. Dichotomous and polytomous scoring are equally welcome.
3. **Existing item analyses.** Any item parameters or calibrations already available for these items, together with the model used. These give us a reference to benchmark our own estimates against.
4. **Linking or equating designs.** Response data connected across forms, groups, or years through common items. Knowing which items are shared is the only addition we need here. This supports our work on extending a scale without recalibration.
5. **Mixed-format assessments.** Tests combining formats, for example selected-response with constructed-response or rubric-scored items, ideally for the same respondents. A format label per item is helpful. These let us place several formats on one scale.
6. **Item content or descriptors (optional).** Item text, or topic and skill classifications, carrying no respondent information. This supports content-based placement of new items and the modeling of skill dimensions.

Scope, format, and governance

A small extract is enough to begin. One subject area, a few thousand respondents, or a limited time window would already let us run a first study. We can work with anonymized or pseudonymized data in any common tabular format, at whatever level of detail and coverage you are comfortable releasing. We are glad to work under a data-sharing or non-disclosure agreement, to restrict use to research, and not to redistribute. A small pilot first, then more if it proves useful, suits us well.